

 ENTERPRISE HYDRAULIC ENGINEERING SUITE

Precision Pump Sizing,

Eliminate disconnected spreadsheets and manual curve interpolation. Select, simulate, and generate publication-ready pump submittal reports in under 3 minutes.

 ISO 9906:2012 ANSI / HI 14.6 API 610 (11th/12th) Darcy-Weisbach Solvers

Engineered for Maximum Workflow Velocity

Built for technical sales teams, consulting hydraulic engineers, and plant operators.



Pump OEMs & Distributors

Accelerate sales quote turnaround from hours to minutes. Equip technical sales representatives with multi-tenant branded catalogs, automated impeller trims, and branded PDF submittal packages.



EPC & Consulting Engineers

Design complete series and parallel pipe systems on an interactive CAD canvas. Calculate friction loss, fitting equivalent lengths, and NPSH margins without fragile Excel formulas.



Mining & Slurry Specialists

Prevent pipe sanding and catastrophic pump wear. Model mixture specific gravities, solids concentrations, Durand deposition limits, and Ferguson & Church settling velocities.

Core Engineering Modules

Version 5.1

Six integrated engines delivering unified hydraulic intelligence.



Duty Point Selection & BEP Analysis

Match target Flow (\$Q\$) and Head (\$H\$) with automatic BEP proximity ranking. Displays Preferred Operating Regions (POR) and Allowable Operating Regions (AOR) to prevent premature bearing and seal failure.



Interactive CAD Pipe Network Designer

Full orthogonal pipe routing with integrated elbows, reducers, and inline valves. Computes Darcy-Weisbach friction head loss using Colebrook-White equations and plots system curves automatically.



Slurry & Viscous Fluid Derating

Native support for abrasive slurries (\$S_m, C_v, V_c\$) and high-viscosity Newtonian fluids according to ANSI/HI standards. Applies automatic \$C_H, C_Q\$, and \$C_{\eta}\$ deratings to pump curves.



High-Precision Curve Digitization

Automated high-order polynomial regression fitting (\$R^2 > 0.998\$). Digitizes test points into smooth H-Q, Efficiency, Power, and NPSHr curves with affinity-law scaling for trimmed impellers and VFD speeds.



Publication-Grade Datasheets & Submittals

Export commercial and technical PDF datasheets with customizable graph layouts (1 to 4 stacked charts), duty point badges, corporate branding, bill of materials, and motor technical notes.



Enterprise Security & Organization Workspaces

Multi-tenant architecture with role-based access control. Proprietary engineering equations execute in a server-side Python calculation vault, keeping formulas and catalogs 100% confidential.

Why Industry Leaders Are Switching

Comparing traditional workflows against Pump Master Pro.

Capability	Legacy Desktop / Excel Spreadsheets	Pump Master Pro Platform
Quotation Preparation Time	2 to 4 hours per project	⚡ Under 3 minutes
Pipe Friction Solving	Simplified Hazen-Williams (water only)	Exact Darcy-Weisbach & Colebrook-White
Slurry & Solids Derating	Manual chart lookups (high risk of error)	Automated Durand V_c & Ferguson-Church V_t
Customer Submittal Deliverables	Fragmented screenshots and printouts	Publication-grade vector PDF datasheets
Deployment & Maintenance	Local installations, DLL conflicts, license keys	100% Cloud SaaS — Instant access on any browser

Ready to Modernize Your Pump Sizing?

Experience automated pump curve fitting, pipe network analysis, and instant submittals.

Explore pumpmasterpro.com →

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